Rigidity of Simplices with Fixed Codimension-2 Face Volumes: Generic and Fold Rigidity, and a Reduction of the Continuous Deformation Conjecture

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Abstract

For $d \geq 4$, the *Continuous Deformation Conjecture* (CDC) asserts that every continuous deformation of a non-degenerate d-simplex in $\mathbb{R}^d$ which preserves the $(d-2)$-dimensional volumes of all its $(d-2)$-faces is a rigid motion. The question was raised by Zhang [6]; it remains open. Let F be the map sending the squared edge lengths of a simplex to the squared volumes of its $(d-2)$-faces, and let Σ be its critical locus. We prove that a non-trivial deformation must lie *entirely* in Σ, and in fact entirely in an explicit real algebraic subset $B \subset \Sigma$ consisting of the non-fold critical points. We deduce that CDC is equivalent to the finiteness of the fibres of $F|_B$, and in particular that CDC follows from quasi-finiteness of F. We explain why quasi-finiteness does not follow from algebraic independence of the face volumes, and why any proof of it must use the Euclidean signature. Finally, we show that the non-congruent pairs of Mohar and Rivin are the two sheets of a fold of F.

1 Introduction

A d-simplex has $\binom{d+1}{2}$ edges and the same number of $(d-2)$-faces. The edge lengths determine the simplex up to congruence. It is natural to ask how much the $(d-2)$-face volumes determine.

Discrete version. The face volumes do *not* determine the simplex, nor even its volume: Mohar and Rivin [4] give, for every $d \geq 4$, pairs of non-congruent d-simplices with the same labelled $(d-2)$-face volumes and different volumes (see Proposition 4.5 and [5]). On the other hand, the $(d-2)$-face volumes are algebraically independent functions of the edge lengths [4], so generically only finitely many simplices share the same face volumes.

Continuous version. The following question was raised by Zhang [6]; we refer to it as the CDC.

Conjecture 1.1 (Continuous Deformation Conjecture, CDC)**.** *Let $d \geq 4$. If $\gamma\colon [0,1] \to (\mathbb{R}^d)^{d+1}$ is a continuous deformation of a non-degenerate d-simplex, through non-degenerate simplices, preserving the $(d-2)$-volumes of all $(d-2)$-faces, then γ is a rigid motion: there are isometries g_t of $\mathbb{R}^d$, depending continuously on t, with $\gamma(t) = g_t\gamma(0)$.*

Zhang [6] shows that the analogue of CDC fails in suitable pseudo-Euclidean spaces; for example, in $\mathbb{R}^{3,1}$ there is a continuous family of non-congruent 4-simplices whose 2-faces lie in Euclidean planes and have constant areas. In Euclidean space the conjecture is open.

Results

We work in the space $X \subset \mathbb{R}^N$, $N = \binom{d+1}{2}$, of squared edge lengths of non-degenerate d-simplices, with the polynomial face map $F\colon X \to \mathbb{R}^N$ and its Jacobian J (Section 2). Let $\Sigma = \{\det J = 0\}$ be the critical locus, and let

$$B = \{x \in \Sigma : \ \nabla(\det J)(x)^T \operatorname{adj} J(x) = 0\}$$

be the set of *non-fold* critical points (Section 5).

Theorem A (Generic rigidity)**.** Σ *is a proper real algebraic subset of* X*. Every* $x \in X \setminus \Sigma$ *is infinitesimally and locally rigid. A continuous cofixed path that is not constant lies entirely in* Σ*.*

Theorem A is close to known results. For $d = 4$, Zhang [6] observes that there are no continuous counterexamples in the generic situation. This follows from Gaifullin's result [2] that generic 2-face areas determine only finitely many congruence classes of 4-simplices. In all dimensions, generic local rigidity follows from the non-singularity of the Jacobian at the regular simplex [4] together with the inverse function theorem. Theorem A makes this precise for all $d \geq 4$ and adds the infinitesimal statement. It also sharpens the conclusion: a non-trivial deformation must lie in Σ at *every* time, not merely meet it. Theorem B builds on this sharpening.

Theorem B (Fold rigidity)**.** *Every fold point of* F *is locally rigid. A continuous cofixed path that is not constant lies entirely in* B*.*

Theorem C (Reduction)**.** *CDC holds if and only if* $F^{-1}(c) \cap B$ *is finite for every* $c \in \mathbb{R}^N$*. In particular, if every fibre of* F *is finite (quasi-finiteness), then CDC holds.*

Theorem C reduces the CDC to a finiteness statement on an explicit real algebraic subset of the critical locus. Section 7 explains why this finiteness is not a formal consequence of algebraic independence, and why any proof of it must use positivity of the Gram matrix.

2 Setup

2.1 Shape space

Label the vertices $0, \ldots, d$ and edges by $e = \{i, j\}$. For $x = (x_e) \in \mathbb{R}^N$ (with $x_{ii} = 0$) define the $d \times d$ matrix

$$G(x)_{ij} = \tfrac{1}{2}\,(x_{0i} + x_{0j} - x_{ij}), \qquad 1 \leq i, j \leq d.$$

If x is the vector of squared edge lengths of a simplex $[v_0, \ldots, v_d]$, then $G(x)$ is the Gram matrix of $v_1 - v_0, \ldots, v_d - v_0$, and $V^2 = \det G(x)/(d!)^2$.

Definition 2.1. The *shape space* is $X = \{x \in \mathbb{R}^N : G(x) \text{ is positive definite}\}$.

X is an open convex cone in $\mathbb{R}^N$, since $x \mapsto G(x)$ is linear and the positive definite matrices form an open convex cone. It is semi-algebraic, being cut out by the leading principal minors of $G(x)$.

Lemma 2.2 (Realisation)**.** *1. A vector* $x \in \mathbb{R}^N$ *is the squared-edge vector of a non-degenerate* d*-simplex in* $\mathbb{R}^d$ *if and only if* $x \in X$*. Two non-degenerate simplices are congruent, by a congruence preserving labels, if and only if they have the same squared-edge vector.*

2. If $x\colon [0,1] \to X$ *is continuous, there is a continuous* $\gamma\colon [0,1] \to (\mathbb{R}^d)^{d+1}$ *such that* $\gamma(t)$ *is a simplex with squared-edge vector* $x(t)$*.*

3. If γ *is a continuous deformation through non-degenerate simplices whose squared-edge vector is constant, then* γ *is a rigid motion in the sense of Conjecture 1.1.*

Proof. (1) If $G(x)$ is positive definite, write $G(x) = R^T R$ with R invertible; then $v_0 = 0$ and $v_i =$ the i-th column of R give $\langle v_i, v_j \rangle = G_{ij}$, hence $\|v_i - v_j\|^2 = G_{ii} + G_{jj} - 2G_{ij} = x_{ij}$, and the simplex is non-degenerate. Conversely a Gram matrix of d linearly independent vectors is positive definite. If two simplices have the same x, the linear map sending $v_i - v_0$ to $v'_i - v'_0$ preserves the Gram matrix, hence is orthogonal, so the simplices are congruent.

(2) Take $R(t)$ to be the Cholesky factor of $G(x(t))$ (upper triangular with positive diagonal), which depends continuously on G, and define $\gamma(t)$ as in (1).

(3) Let $P(t)$ be the matrix with columns $\gamma_i(t) - \gamma_0(t)$, $i = 1, \dots, d$; it is invertible and continuous in t. Put $L_t = P(t)P(0)^{-1}$ and $g_t(y) = L_t(y - \gamma_0(0)) + \gamma_0(t)$. Since $P(t)^T P(t) = G(x) = P(0)^T P(0)$, L_t is orthogonal, so g_t is an isometry, it depends continuously on t, and $g_t\gamma(0) = \gamma(t)$. □

2.2 The face map

For $I \subset \{0, \dots, d\}$ with $|I| = d - 1$, let $f_I(x)$ denote the squared $(d-2)$-volume of the face spanned by $\{v_i : i \in I\}$. By the Gram formula applied to the face, f_I is a polynomial in the x_e with $e \subset I$, homogeneous of degree $d-2$, and $f_I > 0$ on X.

Definition 2.3. The *face map* is $F = (f_I)_I \colon X \to \mathbb{R}^N$. Its Jacobian is $J(x) = \big(\partial f_I / \partial x_e\big)_{I,e}$, an $N \times N$ matrix with polynomial entries, and its *critical locus* is $\Sigma = \{x \in X : \det J(x) = 0\}$.

Using squared volumes instead of volumes changes J by multiplication with invertible diagonal matrices on both sides, so Σ and all statements below are unaffected.

Definition 2.4. A continuous path $x \colon [0,1] \to X$ is *cofixed* if $F(x(t))$ is constant.

By Lemma 2.2, CDC is equivalent to the statement: *every continuous cofixed path in X is constant.* Indeed, a cofixed deformation γ gives a continuous cofixed path $x(t)$ of squared edge lengths; if x is constant, γ is a rigid motion by Lemma 2.2(3). Conversely, a non-constant cofixed path lifts by Lemma 2.2(2) to a deformation which is not a rigid motion. We work with paths in X from now on.

3 The Jacobian at the regular simplex

Let $\mathbf{1} \in X$ denote the regular simplex with unit edges, and let A be the $N \times N$ incidence matrix with $A_{I,e} = 1$ if $e \subset I$ and 0 otherwise.

Proposition 3.1. *$J(\mathbf{1}) = c_d\, A$ with $c_d = \dfrac{1}{((d-2)!)^2\, 2^{d-3}} > 0$, and $\det J(\mathbf{1}) \neq 0$ for $d \geq 3$. More precisely, the eigenvalues of A, after reindexing its rows, are $\binom{d-1}{2}$ (multiplicity 1), $-(d-2)$ (multiplicity d) and 1 (multiplicity $\frac{1}{2}(d+1)(d-2)$), so $|\det A| = \binom{d-1}{2}(d-2)^d$.*

Proof. If $e \not\subset I$ then f_I does not depend on x_e, so $J_{I,e} = 0$. The symmetric group S_{d+1} fixes $\mathbf{1}$ and acts transitively on pairs (e, I) with $e \subset I$, so all such entries of $J(\mathbf{1})$ are equal to a common value c_d. By Euler's theorem for the degree-$(d-2)$ homogeneous polynomial f_I,

$$\sum_{e \subset I} x_e \frac{\partial f_I}{\partial x_e} = (d-2)\, f_I.$$

At $x = \mathbf{1}$ the face is regular with $f_I = \frac{d-1}{((d-2)!)^2\, 2^{d-2}}$, and I contains $\binom{d-1}{2}$ edges, so $c_d\binom{d-1}{2} = (d-2)f_I$, which gives the stated c_d.

For the eigenvalues, index each face I by its complementary pair $\bar{I} = \{0, \dots, d\} \setminus I$. Then $e \subset I$ if and only if $e \cap \bar{I} = \emptyset$, so after reindexing, A is the adjacency matrix of the Kneser graph $K(d+1, 2)$. Its eigenvalues are $(-1)^j\binom{d+1-2-j}{2-j}$ with multiplicity $\binom{d+1}{j} - \binom{d+1}{j-1}$ for $j = 0, 1, 2$ [3, Ch. 9], which are the stated values; all are non-zero for $d \geq 3$. This is also proved in [4]. □

For $d = 4, 5$ this gives $|\det A| = 48$ and 1458, which we also confirmed numerically.

Corollary 3.2. *Σ is a proper real algebraic subset of X. It is closed, nowhere dense and of Lebesgue measure zero.*

Proof. $\det J$ is a polynomial on $\mathbb{R}^N$ that is non-zero at $\mathbf{1}$, and X is a connected open set. A polynomial that is not identically zero has a closed, nowhere dense zero set of measure zero. □

4 Rigidity away from Σ

Proposition 4.1 (Local rigidity off Σ)**.** *If $x \in X \setminus \Sigma$, then F restricts to a diffeomorphism from a neighbourhood W of x onto an open set. In particular, x is the only point of W with face volumes $F(x)$.*

Proof. The inverse function theorem. □

For the infinitesimal statement, let $E\colon (\mathbb{R}^d)^{d+1} \to \mathbb{R}^N$, $E(p)_{ij} = \|p_i - p_j\|^2$, be the edge map. An *infinitesimal motion* of $p = (p_0, \dots, p_d)$ is a vector $\dot p \in (\mathbb{R}^d)^{d+1}$. It is *trivial* if there are a skew-symmetric matrix S and a vector b with $\dot p_i = S p_i + b$ for all i, i.e. if it is the velocity of a path of isometries.

Corollary 4.2 (Infinitesimal rigidity off Σ)**.** *Let p be a non-degenerate simplex with $x = E(p) \notin \Sigma$. If $\dot p$ preserves all $(d-2)$-face volumes to first order, i.e. $d(F \circ E)_p(\dot p) = 0$, then $\dot p$ is trivial.*

Proof. By the chain rule, $d(F \circ E)_p(\dot p) = J(x)\, dE_p(\dot p)$. Since $J(x)$ is invertible, $dE_p(\dot p) = 0$, that is,

$$\langle p_i - p_j,\ \dot p_i - \dot p_j \rangle = 0 \qquad \text{for all } i < j.$$

Put $u_i = p_i - p_0$ and $\dot u_i = \dot p_i - \dot p_0$ for $1 \le i \le d$. The edges $0i$ give $\langle u_i, \dot u_i \rangle = 0$. The edges ij with $i, j \ge 1$ give $\langle u_i - u_j, \dot u_i - \dot u_j \rangle = 0$, and expanding with the previous identities gives $\langle u_i, \dot u_j \rangle + \langle u_j, \dot u_i \rangle = 0$. Let P and $\dot P$ be the $d \times d$ matrices with columns u_i and $\dot u_i$. Then these equations say exactly that

$$P^T \dot P + \dot P^T P = 0.$$

P is invertible because p is non-degenerate. Set $S = \dot P P^{-1}$, so $\dot P = SP$. Then $P^T(S + S^T)P = 0$, hence $S + S^T = 0$ and S is skew-symmetric. Thus $\dot u_i = S u_i$ for all i, and with $b = \dot p_0 - S p_0$ we get $\dot p_i = S p_i + b$ for all i. □

Theorem 4.3 (Theorem A)**.** *Let $x\colon [0,1] \to X$ be a continuous cofixed path. Then either x is constant, or $x(t) \in \Sigma$ for every $t \in [0,1]$.*

Proof. Let $U = \{t : x(t) \notin \Sigma\}$, an open subset of $[0,1]$. If $t_0 \in U$, take W as in Proposition 4.1 for $x(t_0)$. For t near t_0 we have $x(t) \in W$ and $F(x(t)) = F(x(t_0))$, so $x(t) = x(t_0)$. Hence x is locally constant on U, and so constant on each connected component of U.

Suppose $U \neq \emptyset$ and let C be a component of U, on which $x \equiv x^* \notin \Sigma$. If $C \neq [0,1]$, then C has an endpoint $a \in [0,1] \setminus C$, and $a \notin U$ because C is a component. By continuity $x(a) = x^* \notin \Sigma$, so $a \in U$, a contradiction. Hence $C = [0,1]$ and x is constant. □

4.1 Mohar–Rivin pairs

For $t \in \mathbb{R}$ let $T(t) \in \mathbb{R}^N$ be the vector with $x_{01} = t$ and $x_e = 1$ for every other edge e. These are the simplices with one distinguished edge studied in [5], and they give the pairs of [4].

Lemma 4.4. *Let $n \ge 2$ and let $T_n(t)$ be the n-dimensional analogue of $T(t)$. Then $T_n(t)$ is the squared-edge vector of a non-degenerate n-simplex if and only if $0 < t < \frac{2n}{n-1}$, and in that case its squared volume is*

$$V^2 = \frac{t\,\big(2n - (n-1)t\big)}{(n!)^2\, 2^n}.$$

Proof. Compute $G = G(T_n(t))$. We have $G_{11} = t$, $G_{1j} = \frac{1}{2}(t + 1 - 1) = \frac{t}{2}$ for $j \ge 2$, $G_{ii} = 1$ for $i \ge 2$, and $G_{ij} = \frac{1}{2}$ for distinct $i, j \ge 2$. So

$$G = \begin{pmatrix} t & \frac{t}{2}\mathbf{1}^T \\ \frac{t}{2}\mathbf{1} & H \end{pmatrix}, \qquad H = \tfrac{1}{2}\big(I + \mathbf{1}\mathbf{1}^T\big) \in \mathbb{R}^{(n-1)\times(n-1)}.$$

The matrix $\mathbf{1}\mathbf{1}^T$ has eigenvalues $n-1$ (once) and 0, so H is positive definite with $\det H = n/2^{n-1}$ and $H^{-1} = 2\big(I - \frac{1}{n}\mathbf{1}\mathbf{1}^T\big)$. Hence $\mathbf{1}^T H^{-1}\mathbf{1} = \frac{2(n-1)}{n}$, and the Schur complement of H is

$$s(t) = t - \frac{t^2}{4}\cdot\frac{2(n-1)}{n} = \frac{t\,\big(2n-(n-1)t\big)}{2n}.$$

G is positive definite if and only if H is and $s(t) > 0$, i.e. if and only if $0 < t < \frac{2n}{n-1}$. Finally,

$$\det G = \det H \cdot s(t) = \frac{t\,(2n-(n-1)t)}{2^n},$$

and $V^2 = \det G/(n!)^2$. □

Proposition 4.5 (Mohar–Rivin pairs)**.** *Let $d \geq 4$, $t_0 = \frac{d-2}{d-3}$, and*

$$t \in \Big(\frac{4}{(d-1)(d-3)},\ \frac{2d}{d-1}\Big), \qquad t \neq t_0,$$

and put $t' = 2t_0 - t$. Then $T(t), T(t') \in X$ and $F(T(t)) = F(T(t'))$, but

$$V(T(t))^2 - V(T(t'))^2 = \frac{8\,(t_0 - t)}{(d-3)\,(d!)^2\,2^d} \neq 0.$$

In particular, $T(t)$ and $T(t')$ are not congruent.

Proof. Both lie in X. By Lemma 4.4 with $n = d$, we need $0 < t, t' < \frac{2d}{d-1}$. For t this holds by assumption. The condition $t' > 0$ means $t < 2t_0$, which holds because $\frac{2d}{d-1} < \frac{2(d-2)}{d-3}$, i.e. $d^2 - 3d < d^2 - 3d + 2$. The condition $t' < \frac{2d}{d-1}$ means

$$t > 2t_0 - \frac{2d}{d-1} = \frac{2(d-2)(d-1) - 2d(d-3)}{(d-3)(d-1)} = \frac{4}{(d-1)(d-3)}.$$

The interval is non-empty and contains t_0 because $d \geq 4$.

Same face volumes. A face I not containing both 0 and 1 has all edges of squared length 1, so f_I is the same for both. A face containing 0 and 1 has squared-edge vector $T_{d-2}(t)$, so by Lemma 4.4 with $n = d-2$,

$$f_I(T(t)) = \frac{t\,\big(2(d-2) - (d-3)t\big)}{((d-2)!)^2\,2^{d-2}}.$$

This quadratic in t has roots 0 and $2t_0$, so it is invariant under $t \mapsto 2t_0 - t$.

Different volumes. By Lemma 4.4 with $n = d$, $(d!)^2\,2^d\,V^2 = g(t) := 2dt - (d-1)t^2$. Using $t + t' = 2t_0$,

$$g(t) - g(t') = (t-t')\big(2d - (d-1)(t+t')\big) = (t-t')\Big(2d - \frac{2(d-1)(d-2)}{d-3}\Big) = -\frac{4(t-t')}{d-3},$$

and $t - t' = 2(t - t_0)$. Congruent simplices have equal volume. □

This construction is due to Mohar and Rivin [4]. The volume-difference formula and its derivation via the distinguished-edge formula are from [5]. For $d = 4$ the admissible range is $\frac{4}{3} < t < \frac{8}{3}$, and $t = \frac{3}{2}$, $t' = \frac{5}{2}$ give $V^2 = \frac{21}{36864}$ and $\frac{5}{36864}$.

Remark 4.6 (Mohar–Rivin pairs lie off Σ). For $d = 4$, consider the pair $T(3/2)$, $T(5/2)$ of Proposition 4.5. Numerically, the smallest singular values of J at these points are approximately 0.061 and 0.042, so both lie in $X \setminus \Sigma$. Thus F is a local diffeomorphism at each of them but is *not* injective on $X \setminus \Sigma$. Local rigidity is the correct notion; global injectivity off Σ fails.

5 Fold points

5.1 The adjugate identity

Lemma 5.1. *Let $x \in \Sigma$ with $\operatorname{rank} J(x) = N-1$, and let v and ω span $\ker J(x)$ and $\ker J(x)^T$. Then $\operatorname{adj} J(x) = \lambda\, v\, \omega^T$ for some $\lambda \neq 0$, and for every $w \in \mathbb{R}^N$,*

$$d(\det J)_x(w) = \lambda\, \omega^T \big(dJ_x(w)\big)\, v.$$

In particular

$$d(\det J)_x(v) = \lambda\, \mathcal{Q}(x), \qquad \mathcal{Q}(x) := \omega^T D^2 F_x(v,v) = \sum_I \omega_I \sum_{e,e'} \frac{\partial^2 f_I}{\partial x_e \partial x_{e'}}(x)\, v_e v_{e'}.$$

If $\operatorname{rank} J(x) \leq N-2$, then $\operatorname{adj} J(x) = 0$ and $d(\det J)_x = 0$.

Proof. From $J \operatorname{adj} J = \operatorname{adj} J\, J = (\det J) I = 0$, the columns of $\operatorname{adj} J$ lie in $\ker J$ and its rows lie in $\ker J^T$, so $\operatorname{adj} J = \lambda v \omega^T$. Since $\operatorname{rank} J = N-1$, some $(N-1)$-minor is non-zero, so $\lambda \neq 0$. Jacobi's formula gives $d(\det J)_x(w) = \operatorname{tr}\big(\operatorname{adj} J(x)\, dJ_x(w)\big) = \lambda\, \omega^T dJ_x(w) v$. For $w = v$, $(dJ_x(v)v)_I = \sum_{e,e'} \partial^2 f_I / \partial x_e \partial x_{e'}\, v_e v_{e'}$. If $\operatorname{rank} J \leq N-2$, every $(N-1)$-minor vanishes, so $\operatorname{adj} J = 0$ and Jacobi's formula gives $d(\det J)_x = 0$. □

5.2 Fold points and the bad set

Definition 5.2. A point $x \in \Sigma$ is a *fold point* if $\operatorname{rank} J(x) = N-1$ and $d(\det J)_x(v) \neq 0$ for v spanning $\ker J(x)$; equivalently, by Lemma 5.1, $\mathcal{Q}(x) \neq 0$. The *bad set* is $B = \Sigma \setminus \{\text{fold points}\}$.

Lemma 5.3. *$B = \{x \in X : \det J(x) = 0, \quad D(x) = 0\}$ where $D(x) = \nabla(\det J)(x)^T \operatorname{adj} J(x)$, a row vector of polynomials. In particular B is a closed real algebraic subset of X, and $X \setminus B$ is open.*

Proof. Let $x \in \Sigma$. If $\operatorname{rank} J(x) \leq N-2$, then $\operatorname{adj} J(x) = 0$, so $D(x) = 0$ and $x \in B$. If $\operatorname{rank} J(x) = N-1$, then $D(x) = \lambda\, (\nabla \det J \cdot v)\, \omega^T$ with $\lambda \neq 0$ and $\omega \neq 0$, so $D(x) = 0$ if and only if $d(\det J)_x(v) = 0$, i.e. if and only if x is not a fold point. □

Proposition 5.4 (Local rigidity of $F|_\Sigma$ at folds)**.** *Let x^* be a fold point. Then there is a neighbourhood W of x^* such that $\Sigma \cap W$ is a smooth hypersurface and $F|_{\Sigma \cap W}$ is injective.*

Proof. Since $d(\det J)_{x^*}(v) \neq 0$, we have $\nabla \det J(x^*) \neq 0$, so by the implicit function theorem Σ is a smooth hypersurface near x^*, with $T_{x^*}\Sigma = \ker d(\det J)_{x^*}$. This tangent space does not contain v, so $\ker J(x^*) \cap T_{x^*}\Sigma = \operatorname{span}(v) \cap T_{x^*}\Sigma = 0$. Hence $d(F|_\Sigma)_{x^*}$ is injective, i.e. $F|_\Sigma$ is an immersion at x^*, and an immersion is injective on a neighbourhood of the point. □

Theorem 5.5 (Theorem B)**.** *Let $x\colon [0,1] \to X$ be a continuous cofixed path. Then either x is constant, or $x(t) \in B$ for every $t \in [0,1]$. In particular, every fold point is locally rigid.*

Proof. Suppose x is not constant. By Theorem 4.3, $x(t) \in \Sigma$ for all t. Let $U = \{t : x(t) \notin B\}$, open by Lemma 5.3. If $t_0 \in U$, then $x(t_0)$ is a fold point; take W as in Proposition 5.4. For t near t_0, $x(t) \in \Sigma \cap W$ and $F(x(t)) = F(x(t_0))$, so $x(t) = x(t_0)$. Thus x is locally constant on U. The component argument in the proof of Theorem 4.3, with Σ replaced by B, shows that if $U \neq \emptyset$ then x is constant, which is a contradiction. Hence $U = \emptyset$.

For the last statement: if x^* is a fold point, then any cofixed path through x^* is constant, because it does not lie in B. Moreover, x^* is isolated in its fibre. Otherwise points of $F^{-1}(F(x^*))$ accumulate at x^*, and the semi-algebraic set $F^{-1}(F(x^*))$ would contain a non-constant continuous arc ending at x^* [1, Ch. 2]. □

Example 5.6 (The Mohar–Rivin pairs form a fold)**.** Let $d \geq 4$ and $t_0 = \frac{d-2}{d-3}$. By the proof of Proposition 4.5, each $f_I(T(t))$ is either constant in t or symmetric about t_0, so

$$J(T(t_0))\, e_{01} = \frac{d}{dt}\Big|_{t=t_0} F(T(t)) = 0,$$

so $T(t_0) \in \Sigma$ with $e_{01} \in \ker J(T(t_0))$. For $d = 4$ ($t_0 = 2$) we checked numerically that $\operatorname{rank} J(T(2)) = 9$ (the smallest singular value is $\approx 2 \times 10^{-12}$, the next is of order 10^{-1}), and that $\mathcal{Q}(T(2)) \approx 0.104 \neq 0$, consistent with Lemma 5.1. So $T(2)$ is a fold point. The Mohar–Rivin pairs $T(2 \pm s)$ are the two preimages of the fold, which is the local model $(u, s) \mapsto (u, s^2)$.

6 The reduction

Theorem 6.1 (Theorem C)**.** *The following are equivalent.*

1. *CDC holds in dimension d.*
2. *Every continuous cofixed path in X is constant.*
3. *For every $c \in \mathbb{R}^N$, the set $F^{-1}(c) \cap B$ is finite.*

Proof. (1)⇔(2) is Lemma 2.2, as explained after the definition of cofixed paths.

(3)⇒(2): Let x be a non-constant cofixed path. By Theorem 5.5, $x([0,1]) \subset F^{-1}(c) \cap B$ for $c = F(x(0))$. This set is finite, hence discrete, and a continuous map from $[0,1]$ to a discrete space is constant, a contradiction.

(2)⇒(3): The set $S = F^{-1}(c) \cap B$ is semi-algebraic, because X is semi-algebraic, B is algebraic and F is polynomial. If S is infinite, it has dimension ≥ 1. By the cell decomposition theorem [1, Ch. 2] it then contains a semi-algebraic subset homeomorphic to an open interval, which gives a non-constant continuous path in $F^{-1}(c)$, i.e. a non-constant cofixed path. □

Hypothesis 6.2 (QF, quasi-finiteness)**.** *For every $c \in \mathbb{R}^N$, the fibre $F^{-1}(c) \subset X$ is finite.*

Corollary 6.3 (CDC assuming QF)**.** *If Hypothesis 6.2 holds in dimension $d \geq 4$, then CDC holds in dimension d. That is, every continuous deformation of a non-degenerate d-simplex in $\mathbb{R}^d$ preserving all $(d-2)$-face volumes is a rigid motion.*

Proof. QF implies condition (3) of Theorem 6.1, since $F^{-1}(c) \cap B \subset F^{-1}(c)$. Alternatively, directly: a cofixed path lies in a finite fibre, so it is constant, and Lemma 2.2(3) gives the rigid motion. □

Remark 6.4. Theorem 6.1 shows that the only information about fibres needed for CDC is on B. It is weaker than QF in two ways: away from B fibres may a priori be infinite but are automatically discrete (Theorems 4.3 and 5.5), and only fibres of $F|_B$ are involved.

7 Why quasi-finiteness is not automatic

7.1 Algebraic independence does not imply finite fibres

Mohar and Rivin show that f_I are algebraically independent [4]. This implies that F is dominant and that its *generic* fibre is finite, but not that every fibre is finite. For example, the components of $\phi(x, y) = (x, xy)$ are algebraically independent, yet $\phi^{-1}(0,0)$ is the line $\{x = 0\}$. In particular, an ideal of height h can contain more than h algebraically independent elements: the ideal $(x) \subset \mathbb{R}[x, y]$ has height 1 and contains the algebraically independent elements x and xy.

7.2 Any proof must use the Euclidean signature

The polynomial map F is defined on all of $\mathbb{R}^N$. In the pseudo-Euclidean space $\mathbb{R}^{p,d-p}$ the squared "lengths" $x_e = \langle v_i - v_j, v_i - v_j \rangle$ are still defined, and when a face spans a Euclidean subspace its squared volume is given by the same polynomial $f_I(x)$. Zhang's family in $\mathbb{R}^{3,1}$ [6], in which all 2-faces lie in Euclidean planes and have constant areas, therefore gives a non-constant continuous path in $\mathbb{R}^N \setminus X$ on which F is constant. Hence:

Remark 7.1. Any proof of Hypothesis 6.2, or of condition (3) of Theorem 6.1, must use that $G(x)$ is positive definite. An argument that is purely algebraic in the polynomials f_I, such as one based only on algebraic independence, cannot succeed. It would equally rule out Zhang's pseudo-Euclidean families.

Gaifullin [2] shows that for $d = 4$ the volume is not integral over the ring generated by the squared 2-face areas. Hence the squared edge lengths are not integral over $\mathbb{R}[f_I]$ either, F is not a finite map in the algebraic sense, and finiteness cannot come from integrality in this case.

7.3 Open problems

1. Prove that $F^{-1}(c) \cap B$ is finite for all c (equivalently, CDC), using positivity of $G(x)$.
2. Determine the dimension of B. Show that B is a proper subset of Σ, i.e. that folds are generic on Σ, in general dimension; for $d = 4$, Example 5.6 exhibits a fold point.
3. Describe the corank-≥ 2 locus $\{\operatorname{rank} J \leq N - 2\} \subset B$, and in particular decide whether it meets X.

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